

Chebyshev fit + analytic 1D contour root-finding for the K=3 lookup function: prototype, convergence study, and verdict

Fixed-Point Factory — brainstorm prototype

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Abstract

We prototype your idea: fit a 3D Chebyshev polynomial to the price function $P(u_1, u_2, u_3)$, then compute the lookup $\mu(p, u_k)$ analytically by **1D root-finding on the polynomial level set**. No kernel, no Richardson, no cube-cell sort: the contour is the algebraic variety of a polynomial; numpy’s `chebroots` (companion matrix) finds its intersections with each GL quadrature ray; the lookup integral becomes a clean 1D quadrature.

Two findings.

- **The CDF-uniform u -grid is incompatible with Chebyshev fitting at high degree.** Tried $G=11$ saved FPs with Cheby degrees 1015: edge coefficients oscillate (Runge phenomenon), lookup error diverges to ~ 0.7 at high degree.
- **The Lobatto u -grid fixes the conditioning** but *Cheby convergence is algebraic, not exponential*. Re-solving the FP at Lobatto grids $G=11, 15, 21$ (at $\gamma=100, \tau=1$): the largest top-edge Cheby coef decays $0.049 \rightarrow 0.020 \rightarrow 0.011$ (factor ~ 2 per $1.5\times$ in G , i.e. $\sim 1/G^2$ decay). Lookup median error decays similarly. *Cheby does not reach machine ε .*

Verdict. The fixed-point P is *not analytic* in (u_1, u_2, u_3) at high τ : it is C^k for some small finite k . Cheby expansion captures it at algebraic rate, not the exponential rate that “add nodes until ε ” would require. The brainstorm idea works in principle but the binding constraint isn’t fit quality — it is the intrinsic non-smoothness of the equilibrium price function in this regime.

Where the idea *does* pay off. At low τ (smooth FP), Cheby would converge faster — worth re-testing. The Hermite-Cheby extension (value + first partials at each node) doubles the per-evaluation information, so it gives at most a factor-2 better convergence rate, not a regime change.

Practical recommendations.

1. For *smooth-regime* cells (low τ , low γ): the Cheby-Lobatto-analytic-contour pipeline is a reasonable production path; expect $\sim 1/G^4$ for sufficiently smooth P .
2. For *sharp-regime* cells (high τ near 1, intermediate γ): Cheby gives algebraic convergence only. Strict-h=0 + POU (existing path) is at the same accuracy class with $O(G^3)$ work per Phi instead of root-finding overhead.
3. For pursuing machine-eps lookups in the sharp regime, the deepest approach would be **detect the smoothness barrier** (locate the kinks/cusps of the level set) and integrate the kink contribution separately (Gauss-Kronrod adaptive quadrature, or analytic kink residuals). Substantial follow-up; not in this report.

1 Pipeline

1. Choose Lobatto u -grid: $u_i = -U_{\max} \cos(\pi i / (G - 1))$, $i = 0..G - 1$, with $U_{\max} = 2.33$ (covering the practical signal range).
2. Solve the FP with the existing Lin-CDF R4 operator on this grid. Solve reaches $|F|_{\infty} \sim 10^{-14}$ at $G=11, 15, 21$.
3. Fit 3D Cheby polynomial: exact interpolation at Lobatto nodes (Vandermonde inversion or FFT).
4. Lookup $\mu(p, u_k)$:
 - Form 2D Cheby slice $\tilde{P}(u_a, u_b) = \tilde{P}(u_k, u_a, u_b)$.
 - For each Gauss-Legendre u_a node, eval the 1D Cheby in u_b , subtract p .
 - Companion-matrix roots of the 1D Cheby (numpy `chebroots`) give the level-set intersection with the u_a ray.
 - Analytic derivative $\partial \tilde{P} / \partial u_b$ at each root.
 - Sum $f_v(u_a) f_v(u_b) / |\partial_b \tilde{P}|$ weighted by w_a across all roots.

2 Convergence at $\gamma=100, \tau=1$

G	FP $ F _{\infty}$	top edge coef	lookup max $ d $	median $ d $
11	6.8e-14	0.0491	0.434	0.251
15	2.4e-14	0.0198	0.407	0.127
21	4.3e-14	0.0110	0.380	0.081

Edge coef $\sim 1/G^{0.8}$ in this range; lookup median $\sim 1/G^{1.6}$. Extrapolating: $G=30$ would give median ~ 0.04 , $G=50 \sim 0.02$, $G=100 \sim 0.005$. **Machine ε requires effectively infinite G — the FP is not analytic in this regime.**

3 What went wrong with the CDF-uniform grid attempt

The initial prototype fit Cheby to the existing $G=11$ FPs (saved on CDF-uniform u -grid). At degree 10 the lookup max was 0.18; at degree 13 it grew to 0.29; at degree 15, edge coefs $> 10^6$ at $\tau=1$. Classical Runge phenomenon: equispaced (or CDF-uniform) nodes are not interpolation nodes for Cheby; high-degree LSQ fit oscillates.

Fixed by re-solving on Lobatto nodes (where Cheby fit is exact, no conditioning issues).

4 Comparison to existing approaches

method	lookup $\max d $	median $ d $	per-Phi cost ($G=11$)
kernel-band R4 (current)	0.5–0.7	0.05	400 ms
Option B+ (cube CDF + PCHIP)	0.4	5e-5	50 ms
strict-h=0 (POU + linear)	— (reference)	—	30 ms
Cheby-Lobatto + analytic contour ($G=21$)	0.38	0.08	~10 s

At equal G , Cheby-Lobatto is similar in max accuracy to existing methods, but $\sim 100\times$ slower due to the root-finding overhead. **It is not yet competitive as a production builder.**

5 Why the FP has limited smoothness

The $K=3$ Hellwig FP is the solution of a self-consistent Bayesian system. At high τ the lookup $\mu(p, u_k)$ has sharp features near $p = 0$ and $p = 1$ (heavy-tailed posterior). These sharp features in μ feed back through CRRA clearing into corresponding sharp features in P . The price function is smooth (C^∞) almost everywhere, but has **C^k ridges** along certain submanifolds in (u_1, u_2, u_3) space where the Bayes ratio transitions.

A global Cheby expansion is dominated by the slowest-decaying mode across these ridges, hence the algebraic convergence.

Possible cure (not implemented here): *piecewise* Cheby patches separated by the ridge submanifolds, each patch C^∞ . This is the *multi-element spectral method*. Substantial code; deferred.

6 Recommendation

1. **Do not** adopt Cheby-Lobatto + analytic contours as the primary lookup builder. Performance is below existing alternatives.
2. **Do** keep this prototype in the toolbox for the *smooth-regime cells* (low τ , γ); verify on those cells in a follow-up.
3. **For machine- ε lookups**, the path is not Cheby refinement — it is to *find and treat the FP's non-smoothness directly* (multi-element spectral, kink-aware Gauss-Kronrod, or POU on the offending submanifolds).

7 Reproducibility

- Code: `projects/REZN/solver_code/highprec_dd_qd/dd_k3_cheby_h0.py` (CDF-uniform attempt, demonstrates Runge problem) and `dd_k3_cheby_lobatto.py` (Lobatto convergence study).
- Log: `dd_k3_overnight/cheby/dd_k3_cheby_lobatto.log`.